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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Milk Dragon
Selected Domain	Orbyte Smart SME Challenge: Less Busywork, Better Business?
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	Orbyte Project Commitment Copilot
Name (Leader)	Lui Qiu Yang
Email & Phone (Leader)	luiqiuyang1230@gmail.com ; +6011-1087 1596
University Affiliation(s)	Sunway University
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small fit-out and renovation contractors repeatedly ask whether they can take a new job without compromising existing obligations. Today's cash balance can hide upcoming materials, subcontractor and payroll payments, while customer receipts may arrive later.

Owners and staff gather records, check dates, update spreadsheets and seek approval for each quotation. Manual repetition creates busywork and risks double-counting paid bills or missing shortfalls within a week.

Our project automates this recurring commitment review, making cash pressure and decision evidence visible to the owner. The intended benefit is faster, more consistent review before taking on work. User needs and time savings will be tested.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The Copilot turns a new quotation into an owner approval case. Staff paste text into a three-step form. The nemotron-3.5-lightning-30b-a3b model, accessed through NVIDIA's API, extracts payment dates for human confirmation.

Simulated Orbyte sources supply projects, payables, receivables and payroll. Deterministic logic validates inputs and checks daily cash over 13 weeks. Weekly views retain each week's lowest daily balance. Three scenarios compare the job as planned, if things slip and with a later start; fixed rules check cash buffers and approval thresholds.

AI explains results, risks and missing information without recommending acceptance or rejection. Only the owner can decide, with server-enforced permissions. Approval triggers a simulated e-sign hand-off; the activity log preserves evidence.

One Next.js app and PostgreSQL support the prototype deployed on Replit. Workshop topics applied include GenAI APIs, relational databases and cloud hosting. Business data is synthetic; integrations are simulated. No real payments or financial advice are provided.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Application: One Next.js app handles the interface, server-side validation, daily cash calculations, fixed rules, permissions and activity log. PostgreSQL stores requests, cash events, check results and decisions; Prisma provides database access.

Flow: Quotation > AI date extraction > human confirmation > validated inputs > daily calculations over 13 weeks and scenarios > fixed rules > AI summary > owner decision > simulated e-sign and log.

AI: nemotron-3.5-lightning-30b-a3b via NVIDIA's API, with server-side credentials and structured output validation. AI cannot change figures, rules or status.

Sponsor cloud platform: Replit hosts the deployed application, with PostgreSQL persistence. Orbyte data and e-sign are simulated. Weekly views preserve daily cash lows.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We focus on a renovation contractor's recurring job review, using three-step intake, plain-language results and owner approval. The app was created as a new Next.js project and is deployed on Replit.

We will test with a teammate who did not build the interface and seek feedback from 2–3 SME owners; results remain unconfirmed. Tests cover daily balances, intraweek shortfalls, duplicate or partially paid obligations, unconfirmed AI dates, permissions and AI failures.

Before presentation, we will replay the seeded journey on Replit and retain deployment evidence. Live integrations, monitoring and notifications remain roadmap items.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: Target 8.2, productivity through technology and innovation. Supporting Target 8.3, productive activity and SME growth. The prototype measures review efficiency as a contribution to these aims, not national SDG indicators. Source: UN DESA, Goal 8, https://sdgs.un.org/goals/goal8#targets_and_indicators
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Planning estimate: 30 minutes per manual job check; this is an assumption, not a measured result. Establish the actual baseline using 10 paired synthetic case runs across 2–3 testers, comparing spreadsheet review with the app on the same inputs. Replace the estimate with the measured median.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: Median active minutes per job check, from opening the same source records to recording an owner decision. Target: 30% less time, provisionally 30 to 21 minutes. Include input confirmation and correction time. Guardrails: Correct daily balances and rule findings against expected results; AI dates confirmed by a person; every final decision recorded by the owner.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target: At least 30% less review time versus the measured manual baseline across 10 paired synthetic case runs. Counterbalance test order and report the small sample size. All fixed calculation and rule tests must pass. Environmental savings are not quantified.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js	Code framework	MIT
Prisma	Code ORM	Apache-2.0
PostgreSQL	Database	PostgreSQL License
Nemotron via NVIDIA API	Pre-trained model service	Applicable NVIDIA API terms
Replit	Cloud platform	Replit service terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Yang

24/10/2026